

# Tirth Sidhpura

Software Engineer

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## SUMMARY

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Software engineer with 3 years of experience building scalable, production-grade backend and full stack systems. Strong in distributed systems, cloud infrastructure, and data-intensive applications. Proven ability to ship reliable features end-to-end and operate systems in production.

## EXPERIENCE

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### Software Engineer (Full Stack – React & Backend)

Erience Solutions

Oct 2022 – Oct 2024

- Built and owned a SEBI-compliant Structured Digital Database (SDD) platform over **18+ months** to manage Unpublished Price Sensitive Information (UPSI) across **thousands of organizations**, supporting concurrent multi-user access, append-only audit logs, and point-in-time reconstruction of access history for regulatory inspections.
- Developed React-based frontend modules for compliance and KYC workflows within **6 months**, implementing secure photo and document uploads, schema-driven data capture, and client-side validation aligned with **SEBI audit accuracy and completeness requirements**.
- Designed and evolved reusable React UI components across multiple releases for document preview, upload progress tracking, validation feedback, and role-based visibility, reducing KYC rework cycles and improving turnaround time for compliance and operations teams.
- Implemented backend APIs to validate, store, and version KYC data and documents, enabling approval, rejection, and resubmission workflows with **full historical traceability** and immutable records required for regulatory audits.
- Architected a distributed backend system with service-level RBAC, append-only audit logging, and SEBI-aligned reporting workflows, ensuring **continuous audit-readiness** and zero-loss traceability in production environments.
- Built scalable ingestion and reconciliation pipelines to process **high-volume daily financial records** from CDSL and NSDL, incorporating checksum-based data integrity checks, exception handling, and reconciliation reports to surface and resolve mismatches.
- Automated Registrar and Transfer Agent (RTA) post-trade workflows and periodic regulatory report generation using scheduled backend jobs, significantly reducing manual operational effort and compliance turnaround times.
- Collaborated closely with compliance and operations stakeholders to translate evolving SEBI regulations into end-to-end, production-ready platform features, balancing regulatory interpretation with system design constraints.
- Owned production services by monitoring live ingestion and reconciliation pipelines, diagnosing batch failures and data inconsistencies, and deploying fixes in a **Linux-based production environment** supporting compliance-critical systems.
- Supported CI/CD pipelines and production deployments, ensuring platform stability, controlled releases, and uninterrupted audit-readiness of regulatory systems.

## PROJECTS

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### RSI Trading & Automation Platform (GitHub Repository)

- Built a multithreaded system to compute RSI indicators across 100+ trading pairs simultaneously.
- Implemented automated trading strategies with stop-loss and take-profit logic using Binance APIs.
- Processed real-time market data with low-latency execution and fault-tolerant order handling.

### PaperOnBoard.in

- Built and scaled an educational platform serving over 10,000 daily users.
- Developed backend services using PHP and SQL, managing production traffic and deployments.
- Created and managed social media channels, driving consistent organic user growth.

### Hybrid Weather Data Engineering & Analytics Platform (GitHub Repository)

- Designed and implemented an end-to-end backend data pipeline ingesting real-time and historical data from multiple APIs.
- Used Node.js and Apache Kafka to process asynchronous data streams reliably.
- Stored and queried structured datasets using PostgreSQL and time-series techniques.

## SKILLS

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**Languages** - JavaScript, TypeScript, Python, SQL

**Frontend** - EJS, HTML, CSS, Bootstrap

**Backend & Systems** - Node.js, Express.js, REST APIs, distributed systems, event-driven architecture

**Cloud & DevOps** - AWS (EC2, S3), Microsoft Azure, DigitalOcean, Docker, CI/CD pipelines

**Messaging & Streaming** - Apache Kafka, message queues, event processing

**Data & Analytics** - PostgreSQL, MySQL, data pipelines, performance tuning, Apache Superset

**Security** - JWT authentication, role-based access control (RBAC), data encryption

**Monitoring & Reliability** - Logging, error tracking, production debugging

**EDUCATION**

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**Master of Computer Science** - Algoma University, Canada